

Introduction to Machine Learning

Lesson Plan

1. What is ML?
2. Why ML?
3. ML workflow
4. ML paradigms
5. Tools
 - Python Jupyter notebooks / Google Colab

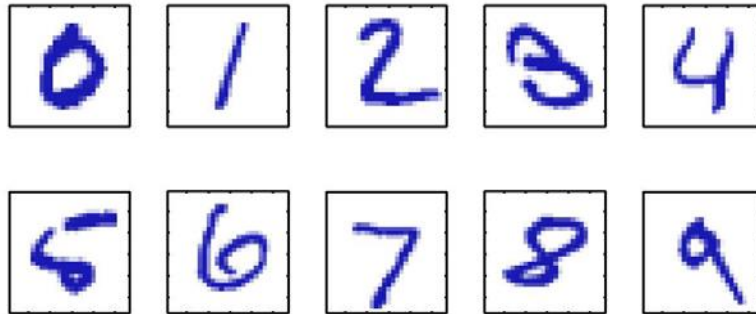
What is ML ?

Machine learning examples

- ❑ **AlphaGo Zero:** Learning to play Go all by itself
- ❑ **Google Translate:** Learning to translate texts from one language to another
- ❑ **Google Voice:** Learning to recognise voice (speech to text)
- ❑ **Amazon Echo / Alexa:** Learning to follow orders
- ❑ **IBM Watson:** Learning to ask and answer questions
- ❑ and many more ...

What is machine learning?

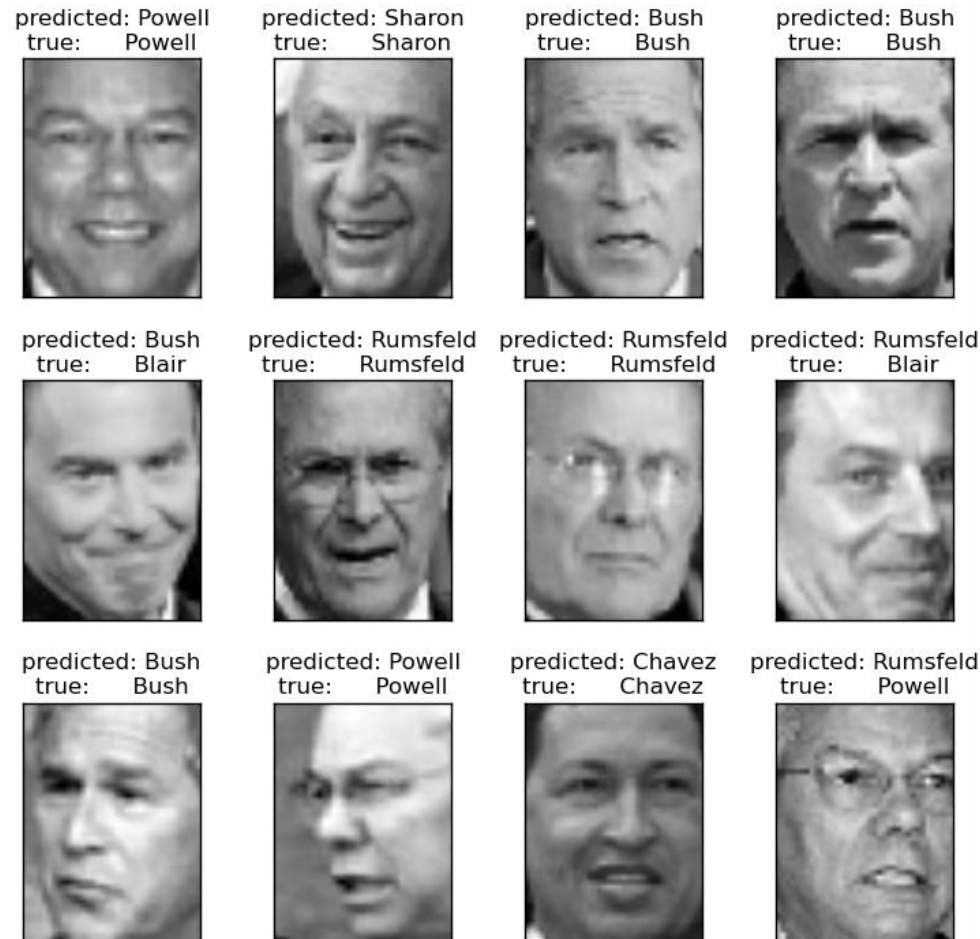
□ Handwritten characters recognition



MNIST handwritten digits dataset
- standard for tests in machine learning.

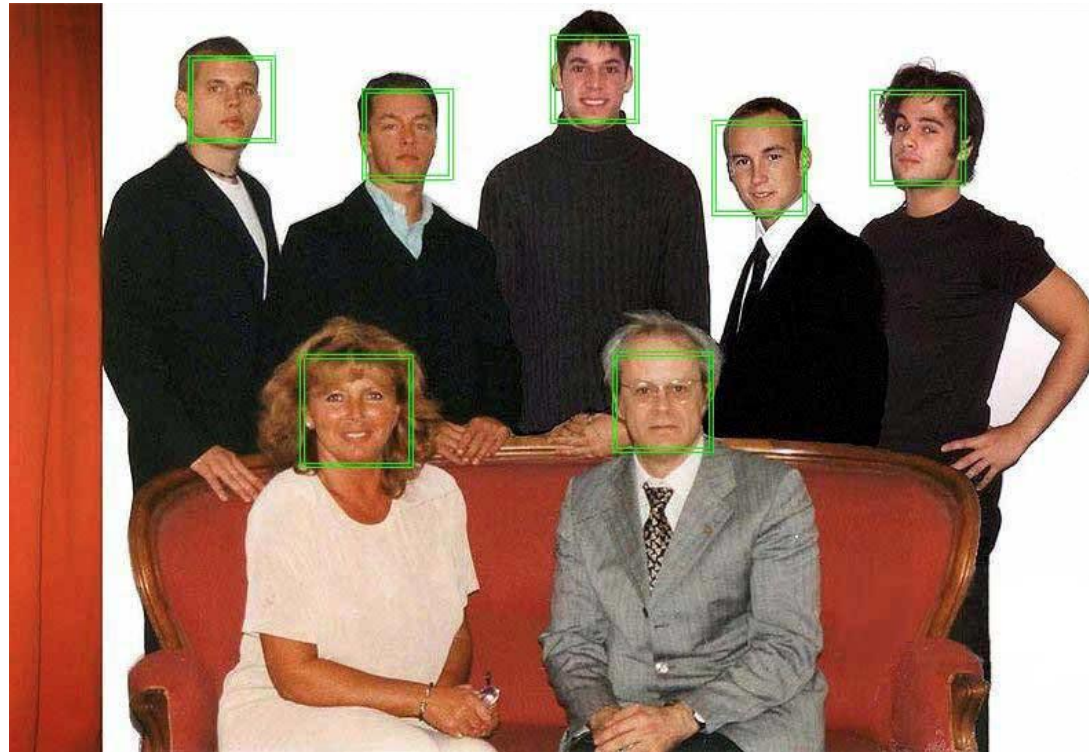
What is machine learning?

□ Facial recognition – whose face is it?



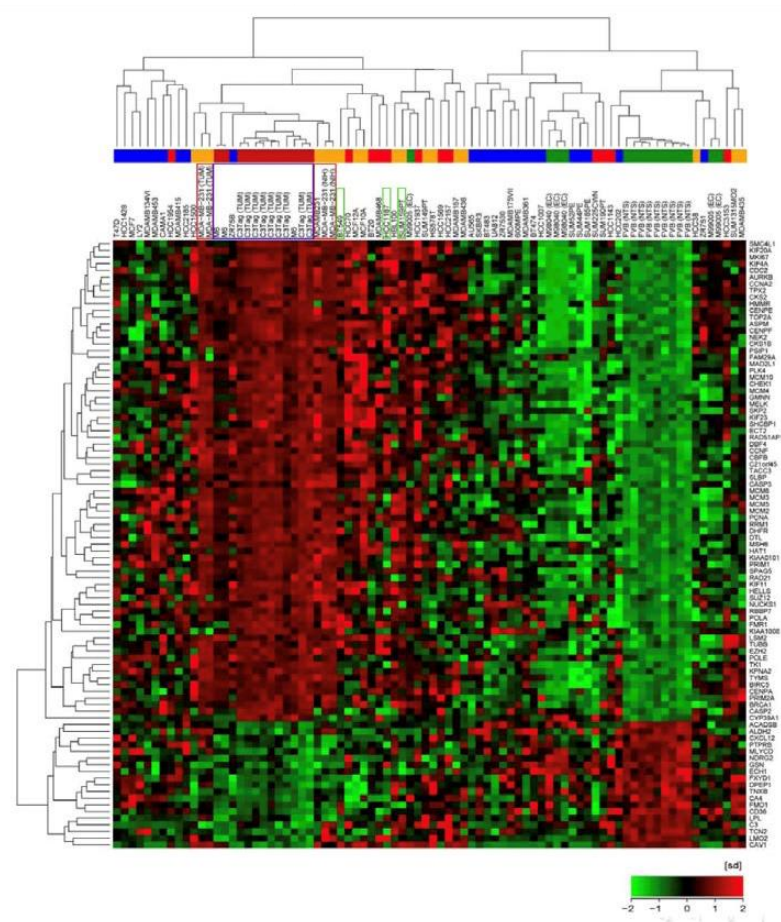
What is machine learning?

❑ Facial detection – where are the faces?



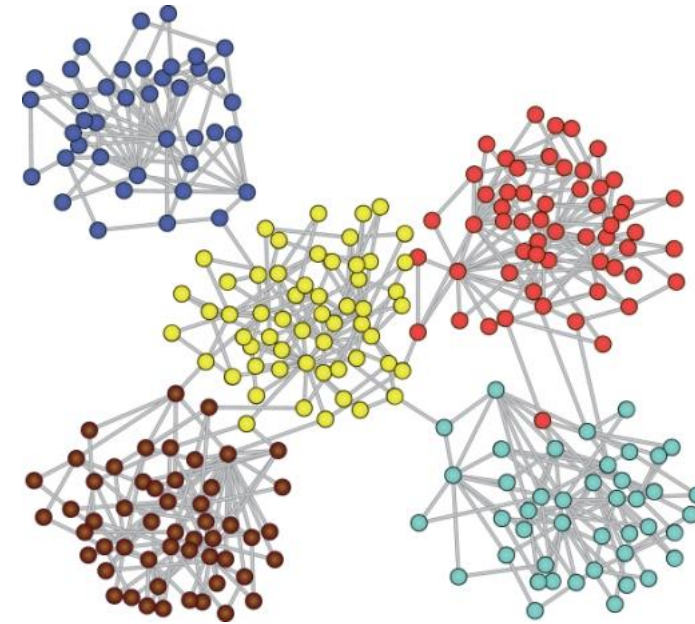
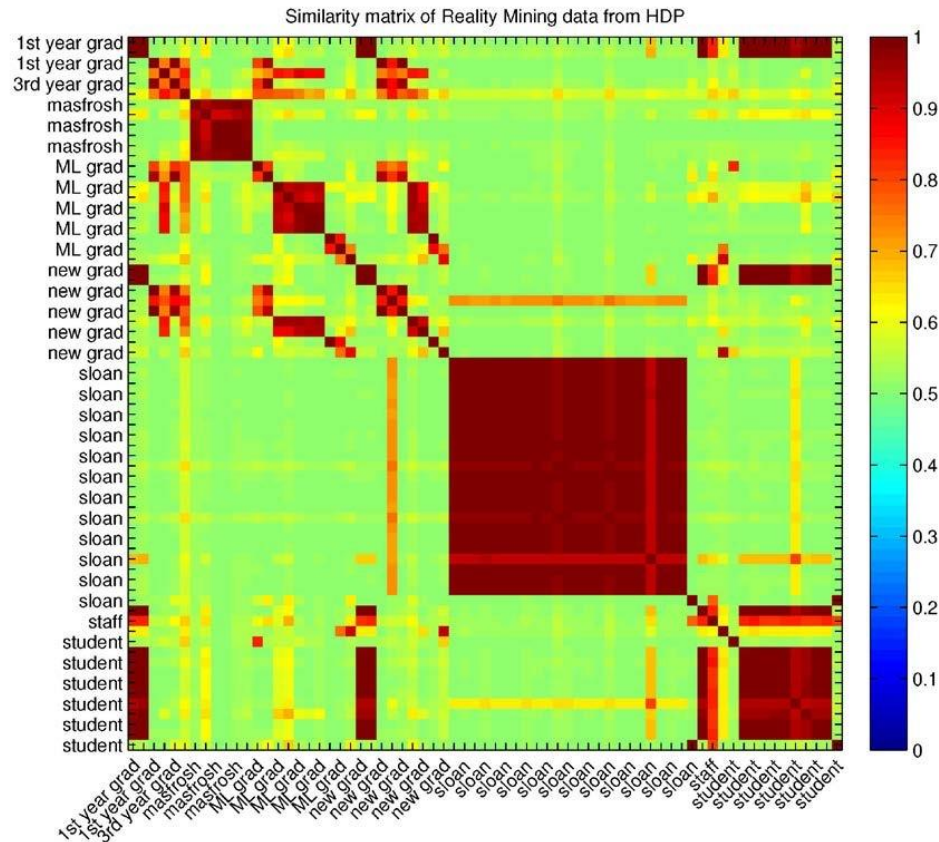
What is machine learning?

- ❑ Gene expression detection – what are the gene expression profiles?



What is machine learning?

- ❑ Community detection – who are the groups? How they are connected within and between them?



What is machine learning?



Deep Blue vs. Kasparov chess matches



Deep Blue
IBM chess computer

Garry Kasparov
World Chess Champion

First match

- February 10, 1996: takes place in Philadelphia, Pennsylvania
- Result: **Kasparov**–Deep Blue (4–2)
- Record set: First computer program to defeat a world champion in a *classical game* under tournament regulations

Second match (rematch)

- May 11, 1997: held in New York City, New York
- Result: **Deep Blue**–Kasparov (3½–2½)
- Record set: First computer program to defeat a world champion in a *match* under tournament regulations

Stanley (vehicle)

From Wikipedia, the free encyclopedia

Stanley is an [autonomous car](#) created by [Stanford University's](#) Stanford Racing Team in cooperation with the [Volkswagen Electronics Research Laboratory \(ERL\)](#). It competed in, and won, the [2005 DARPA Grand Challenge](#),^[1] earning the Stanford Racing Team the 2 million dollar prize.



Stanley parked after the 2005 DARPA Grand Challenge

Watson (computer)

From Wikipedia, the free encyclopedia

"IBM Watson" redirects here. For the laboratory, see [Thomas J. Watson](#)

Watson is an [artificially intelligent](#) computer system capable of answering questions posed in [natural language](#),^[2] developed in IBM's DeepQA project by a research team led by [principal investigator David Ferrucci](#). Watson was named after IBM's [Thomas J. Watson](#).^{[3][4]} The computer system was specifically developed to answer questions on the [quiz show Jeopardy!](#).^[5] In 2011, Watson competed on *Jeopardy!* against former winners [Brad Rutter](#) and [Ken Jennings](#).^{[3][6]} Watson received the first prize of \$1 million.^[7]



Watson, Ken Jennings, and Brad Rutter in their *Jeopardy!* exhibition match.

Superhuman performance!

What is machine learning?

- ❑ *“Field of study that gives computers the ability to learn without being explicitly programmed”, Arthur Samuel.*

In Memoriam

Arthur Samuel: Pioneer in Machine Learning

Arthur Samuel (1901–1990) was a pioneer of artificial intelligence research. From 1949 through the late 1960s, he did the best work in making computers learn from their experience. His vehicle for this work was the game of checkers.

Programs for playing games often fill the role in artificial intelligence research that the fruit fly *Drosophila* plays in genetics. *Drosophila* are convenient for genetics because they breed fast and are cheap to keep, and games are convenient for artificial intelligence because it is easy to compare a computer's performance on games with that of a person.

Samuel was a modest man, and the importance of his work was widely recognized only after his retirement from IBM in 1966, in part because he didn't relish the politics that were required to have his research more vigorously followed up on. He was also realistic about the large difference between what had been accomplished in understanding intellectual mechanisms and what would be required to reach human-level intelligence.

Samuel's papers on machine learning are still worth studying. Possessing great creativity and essentially working alone, doing his own programming, he invented several seminal techniques in rote learning and generalization learning, using such underlying techniques as mutable

strate the power of electronic computers. He didn't finish the

program while he was at the university of Illinois, perhaps because the computer wasn't finished in time.

In 1949, Samuel joined IBM's Poughkeepsie Laboratory, where he worked on IBM's first stored program computer, the 701. This computer's memory was based on Williams tubes, which stored bits as charged spots on the screen of a cathode ray tube. Samuel managed to increase the number of bits stored from the customary 512 to 2048 and to raise the



What is machine learning?

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- ❑ *“A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E ”, Tom Mitchell.*

What is machine learning?

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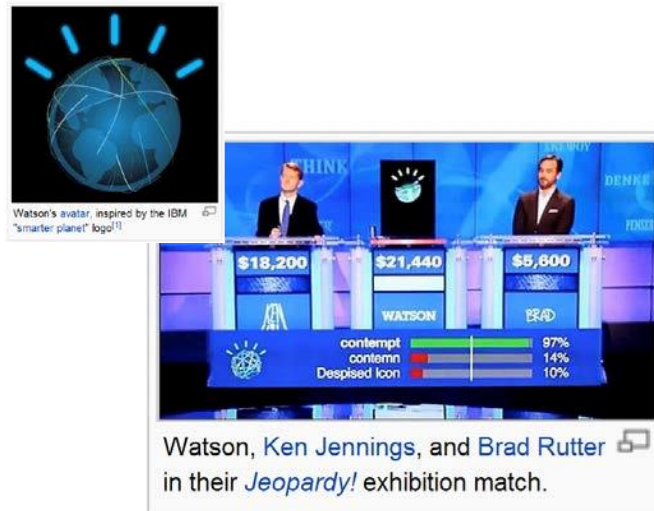
□ **Learning = improving performance at some task with experience.**

- Improve over task T
- With respect to performance measure P
- Using experience E

What is machine learning?

- “A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E ”, Tom Mitchell.

IBM Watson



Experience E	Task T	Performance P
Databases of millions of question-answer pairs	Given a question, find the best answer	How accurate the answer is

What is machine learning?

Facial recognition

predicted: Powell
true: Powell



predicted: Sharon
true: Sharon



predicted: Bush
true: Bush



predicted: Bush
true: Bush



predicted: Bush
true: Blair



predicted: Rumsfeld
true: Rumsfeld



predicted: Rumsfeld
true: Rumsfeld



predicted: Rumsfeld
true: Blair



predicted: Bush
true: Bush



predicted: Powell
true: Powell



predicted: Chavez
true: Chavez



predicted: Rumsfeld
true: Powell



Experience **E**

Databases of
thousands of known
faces

Task **T**

Given a new photo,
recognize the name
of the face

Performance **P**

How accurate the
recognition is



What is machine learning?

- “A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E ”, Tom Mitchell.

Detecting Spam Emails:

Experience E	Task T	Performance P
Examples of spam emails and non-spam emails	???	???

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Detecting Spam Emails:

Experience E	Task T	Performance P
Examples of spam emails and non-spam emails	To assign a label 'spam' or 'non-spam' to an email	How accurate is detection of spam email

What is machine learning?

- “A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E ”, Tom Mitchell.

Detecting credit card fraud:

Experience E	Task T	Performance P
???	???	???

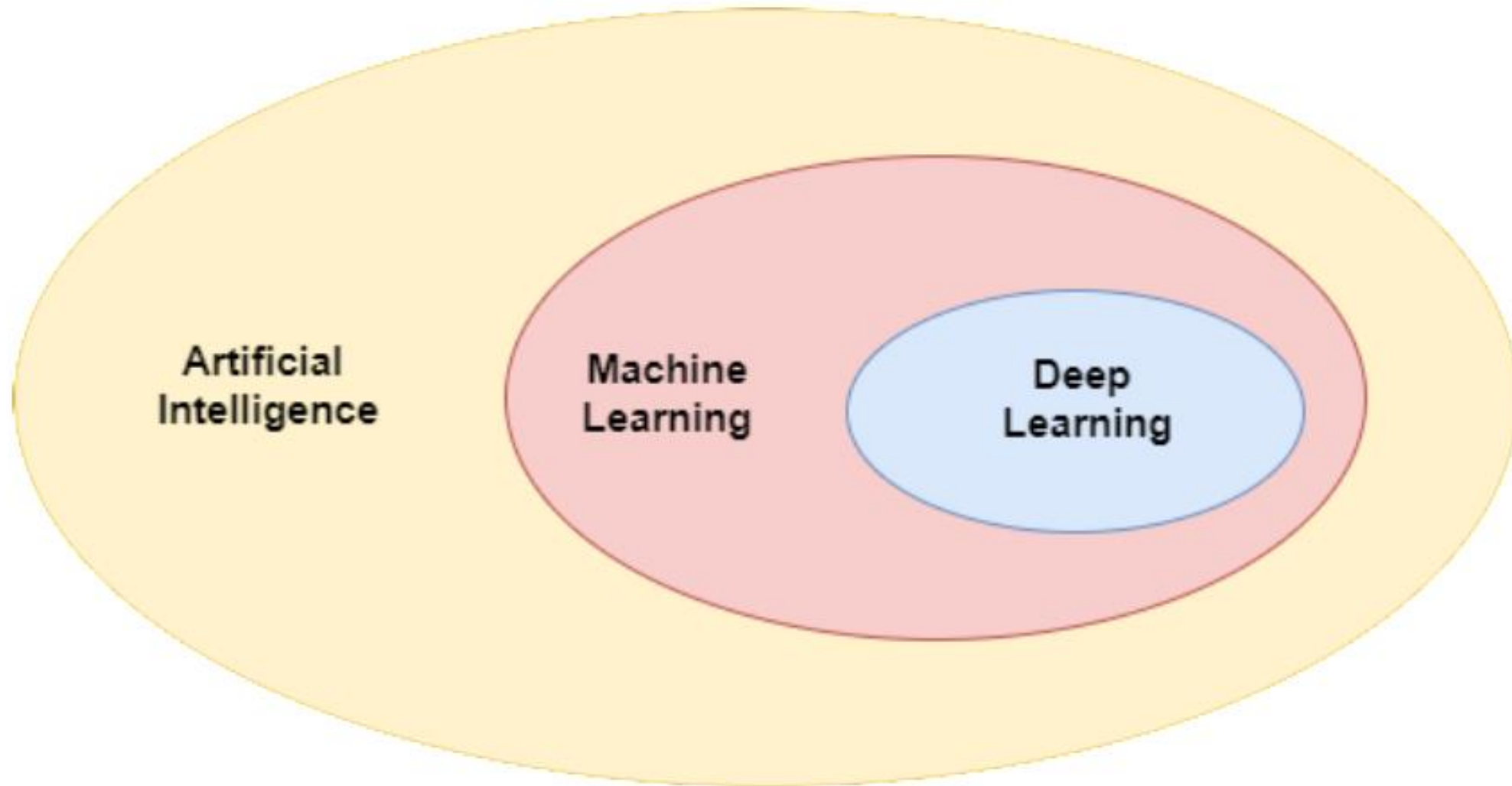
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Detecting credit card fraud:

Experience E	Task T	Performance P
Data collected for credit card transactions deemed as fraud and non-fraud	Assign a label ‘fraud’ or ‘non-fraud’ to a given credit card transaction	How accurate is detection of credit-card fraudulent transaction

What is not ML ?

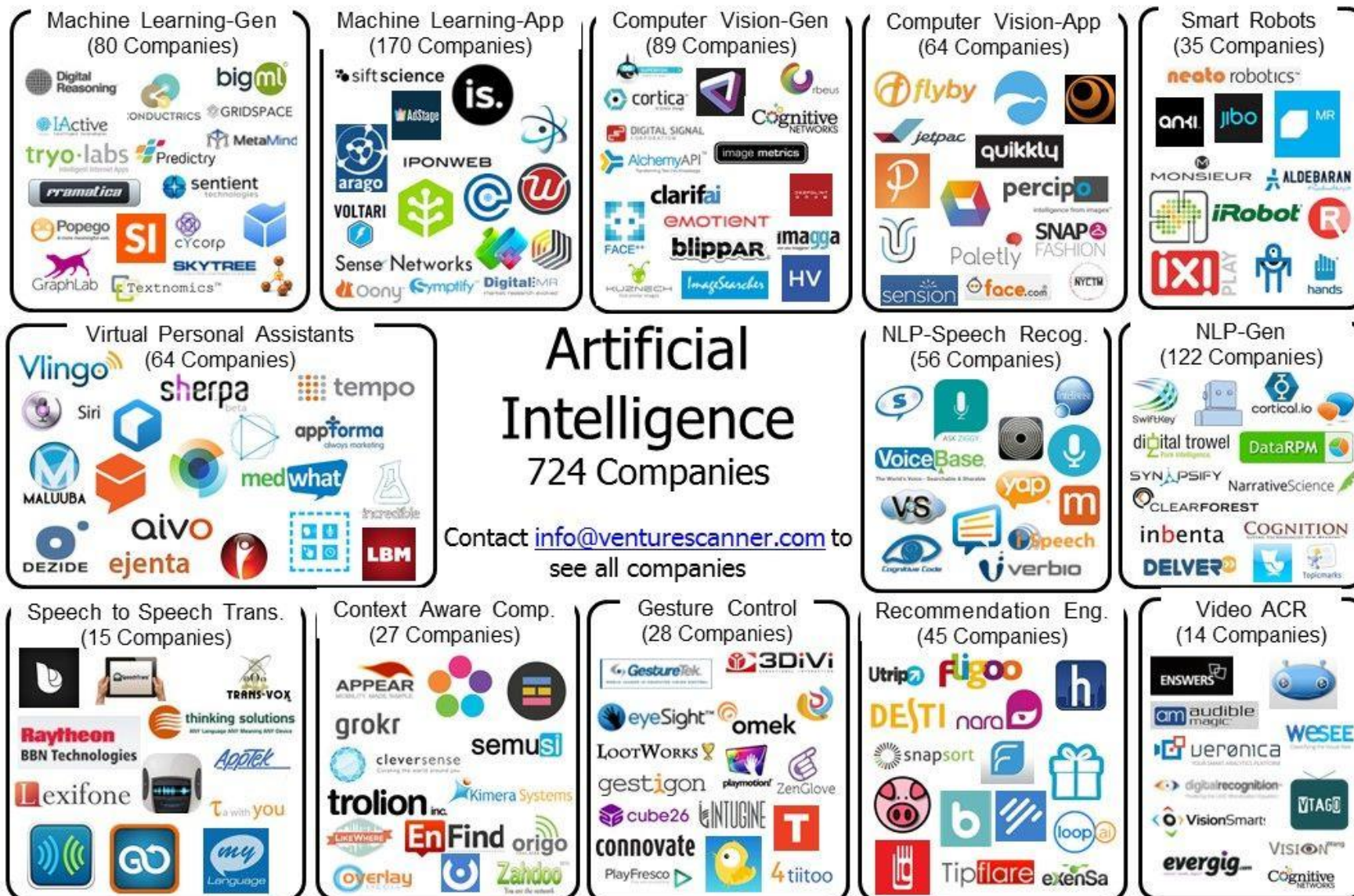


Why ML?

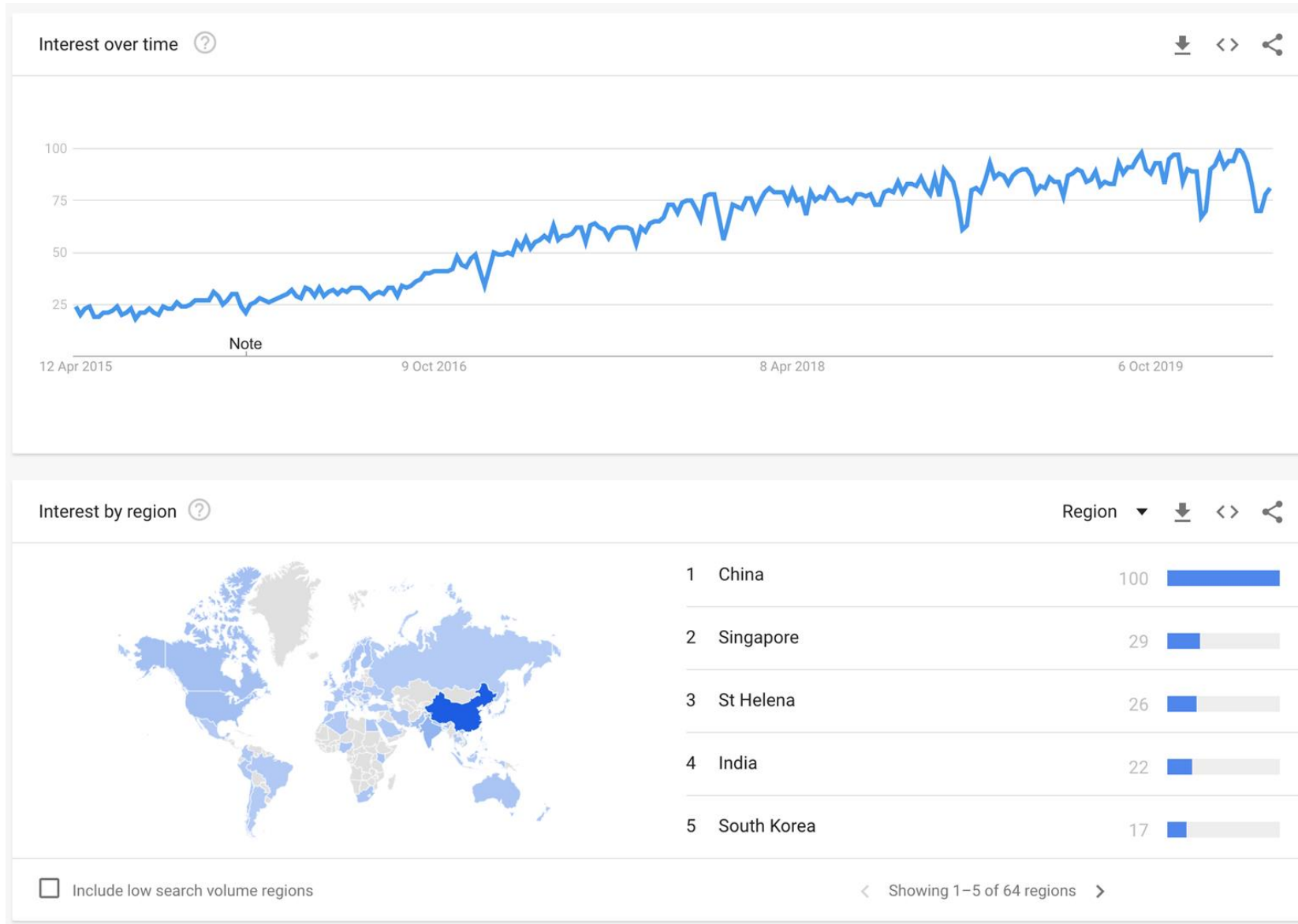
Why machine learning?

- ❑ Growing flood of data – big data, which is impossible to analyse by humans.
 - Trillion web pages
 - Trillions emails exchanged per day, etc ...
- ❑ Computational power is widely available
- ❑ Lots of progress in algorithms and theory – since 1950s!
- ❑ Explosive interest of industry, startups, investors and govt
- ❑ Solving problems of societal impacts, humanity and nature.

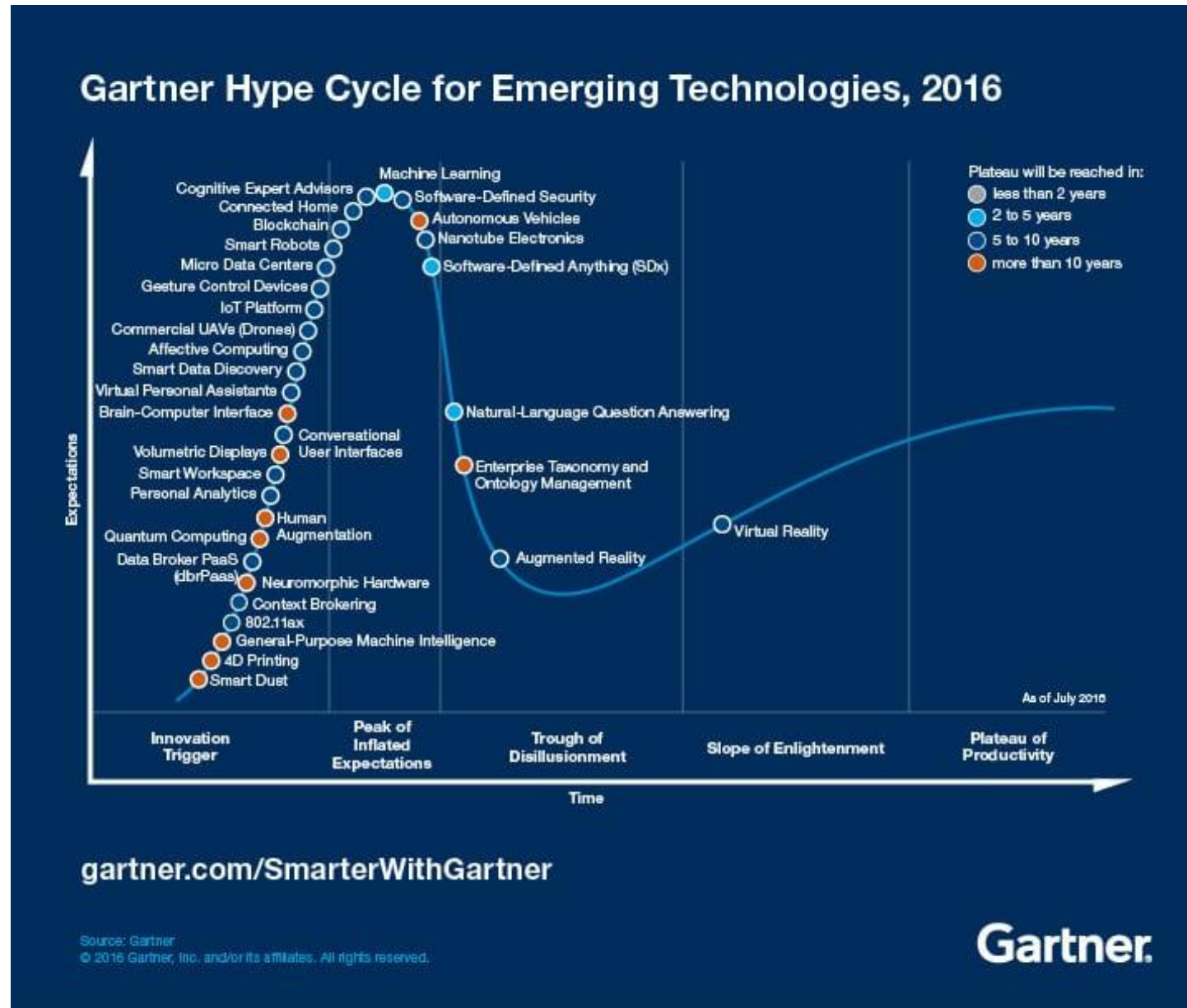
Why machine learning?



Why machine learning?



Why machine learning?

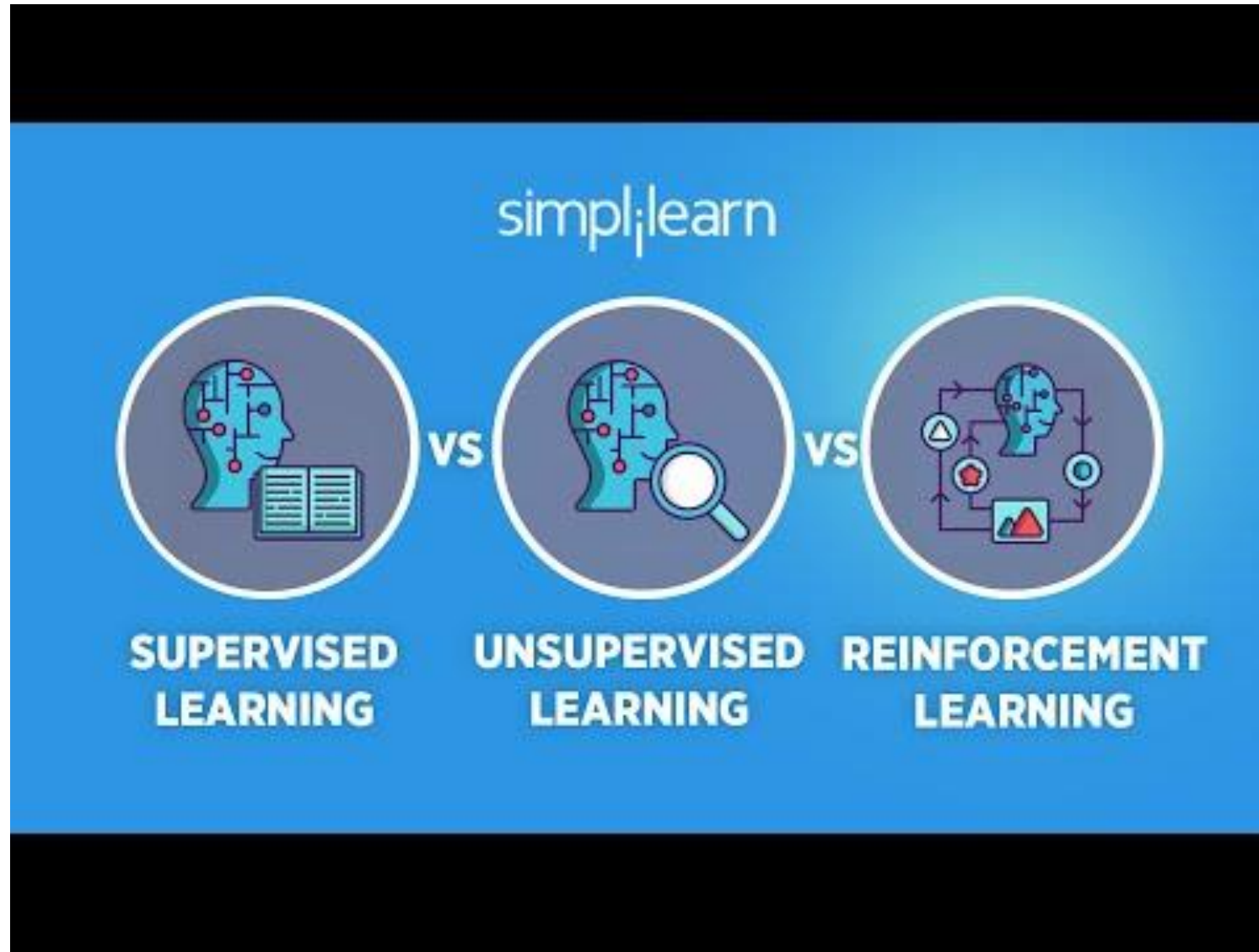


ML Workflow

The Machine Learning Workflow



ML Paradigms



Some useful tools

A visual introduction to ML

R2
D3

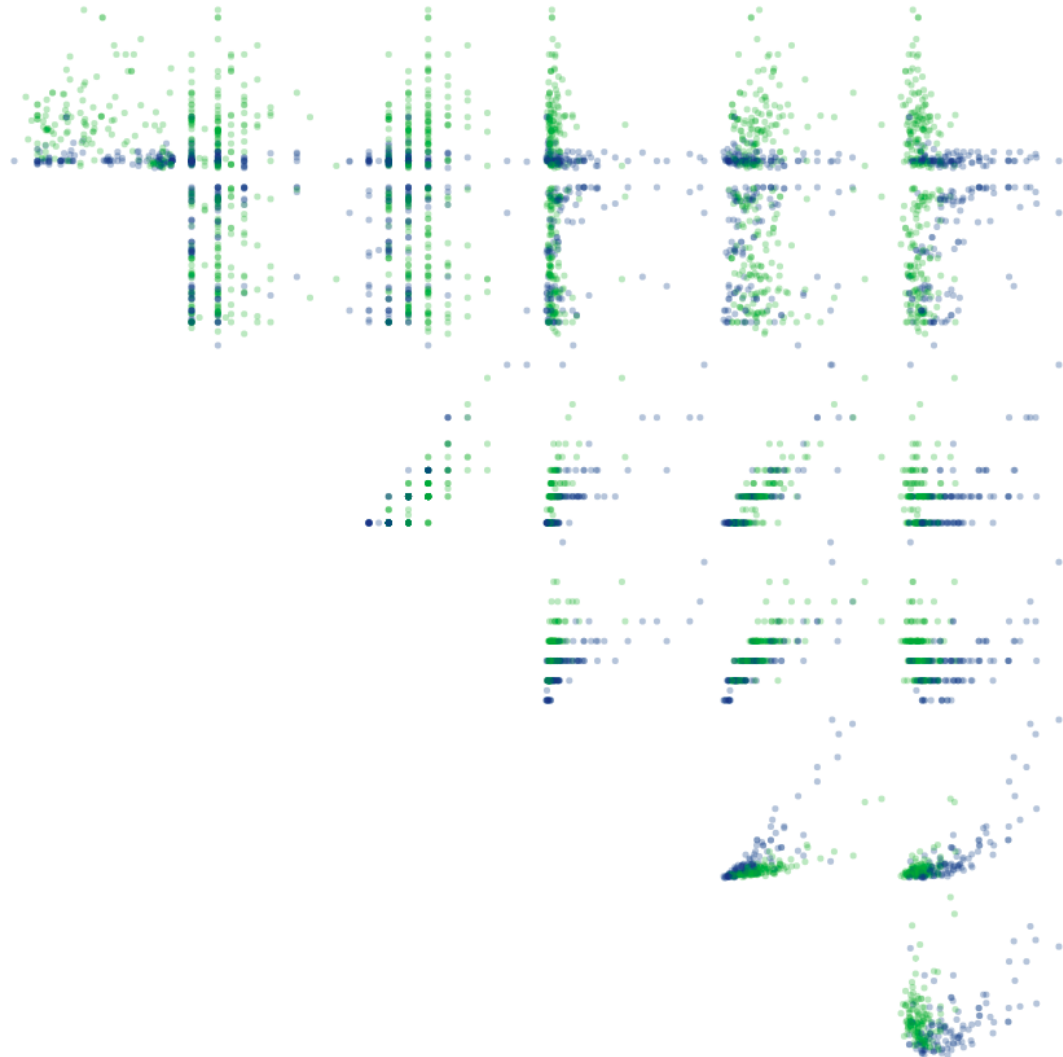
A visual introduction to machine learning

🌐 English ▾

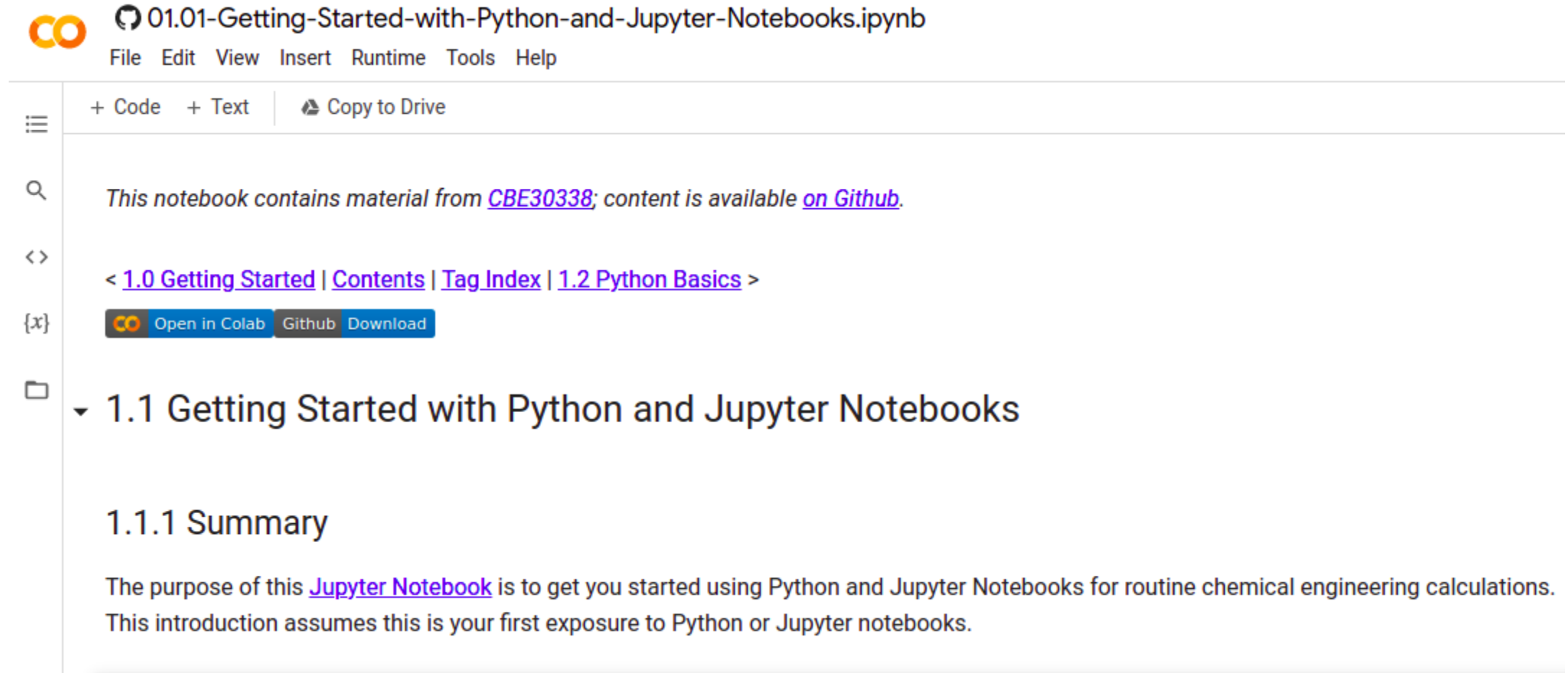
In machine learning, computers apply **statistical learning** techniques to automatically identify patterns in data. These techniques can be used to make highly accurate predictions.

Keep scrolling. Using a data set about homes, we will create a machine learning model to distinguish homes in New York from homes in San Francisco.

SCROLL



Google Colab / Jupyter Notebooks



The screenshot displays the Google Colab interface for a Jupyter Notebook. At the top, the notebook title is "01.01-Getting-Started-with-Python-and-Jupyter-Notebooks.ipynb". Below the title is a menu bar with options: File, Edit, View, Insert, Runtime, Tools, and Help. A toolbar shows "+ Code", "+ Text", and "Copy to Drive". On the left sidebar, there are icons for a menu, search, code editor, variable explorer, and file explorer. The main content area contains the following text:

This notebook contains material from [CBE30338](#); content is available [on Github](#).

< [1.0 Getting Started](#) | [Contents](#) | [Tag Index](#) | [1.2 Python Basics](#) >

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▼ 1.1 Getting Started with Python and Jupyter Notebooks

1.1.1 Summary

The purpose of this [Jupyter Notebook](#) is to get you started using Python and Jupyter Notebooks for routine chemical engineering calculations. This introduction assumes this is your first exposure to Python or Jupyter notebooks.